OFFICIAL DOCUMENTATION & USER MANUAL

TC OpenSCAD Card Script Generator

Interactive design workstation for flush two-colour smart cards, NFC/QR badges, and asset tags on the Bambu Lab P2S (256 × 256 mm). Web landing page: https://drtancheng.github.io/card-generator/

0.20 mm	1 Swap	≥ 0.80 mm
FLUSH INLAY FIRST LAYER	SINGLE COLOR TRANSITION	RELIABLE STROKE TARGET

Core Architecture & Standalone Advantages

The TC Card Generator provides a self-contained executable that eliminates Python environment setup, command-line dependencies, or manual OpenSCAD code construction. Everything required for typography calculation, vector layout, QR compilation, and CV print validation runs locally on your workstation.

No Environment Setup

Bundled standalone Windows executable. Zero installation of Python runtimes, Tkinter libraries, Pillow, or QRCode dependencies needed.

Flush Inlay Manufacturing

Text and QR modules form the bottom 0.20 mm of the model printed face-down on the build plate. No dust-trapping grooves or raised letters.

Nozzle-Physics Engine

Real-time stroke thickness calculation compares font outlines to your 0.40 mm nozzle, flagging lines that risk failing extrusion.

Simulated Optical Scanner

Integrated computer-vision model tests QR codes against physical over- and under-extrusion blur to guarantee camera scannability.

Why Flush Inlay Over Raised / Recessed Tags

Standard 3D printed tags rely on raised embossing or recessed debossing. Both compromise utility: raised characters snap off or scuff against surfaces, while recessed cavities collect dirt, grease, and shadows that render phone cameras unable to read QR patterns. By printing flush face-down on a textured or smooth PEI sheet, the finished part exhibits an injection-molded, glass-smooth exterior.

Interactive Canvas & Customization Mechanics

Every element on the card can be modified via the parameter controls on the left panel or manipulated directly on the interactive 2D canvas with keyboard and mouse actions.

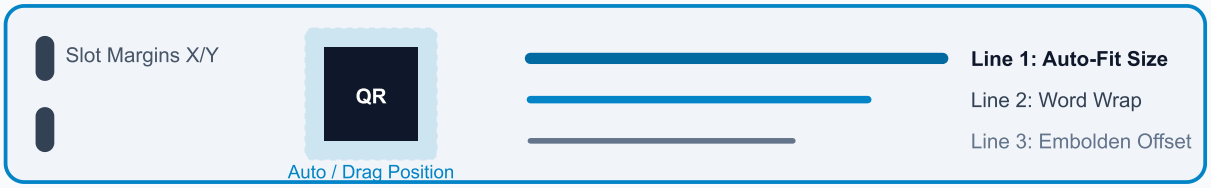


Figure 1: Spatial layout zones for slots, QR quiet zones, and auto-wrapped typography.

Component Modification Guide

Component	Control Methods	Physical Print Rules & Safe Limits
Plate Dimensions	Presets (Credit, A7, A6) or custom W × L inputs.	Max 256 × 256 mm (P2S Bed). Default thickness 1.8 mm.
Zip-Tie Slots	13 placement modes (Corners, Centers), Len, Width.	Maintain ≥ 1.2 mm wall margin to card edge to avoid breakout.
QR Code Positioning	Position menu (Top/Bottom/Left/Right) or click-drag.	Keep 4-module quiet zone buffer around ink to ensure phone detection.
QR Sizing & Modules	“Fit QR Size” button or mouse-wheel scroll.	Modules must be ≥ 0.8 mm (2× nozzle). ECC L yields larger modules.
Inlay Text & Fonts	1 to 8 lines, system TrueType font picker.	Minimum 1.5 mm height. Stroke width must reach ≥ 0.8 mm.
Auto-Fit & Wrap	Checkboxes: “Auto-fit”, “Auto-wrap long lines”.	Wraps phrases at punctuation and words; shrinks only if blocked.
Embolden Offset	“Make text printable” or manual mm slider.	Expands glyph outline laterally (0.10 mm standard) to close thin gaps.
Nudge & Snap	Arrow keys (0.5 mm), Shift (0.1 mm), Ctrl (2 mm).	Magnetic alignment guides snap automatically to plate and QR centers.

Pro Tip: Click “Make text printable” in Section 4. The tool analyzes all glyphs, determines the ideal embolden value, and scales text automatically so every stroke reliably prints.

Exporting from the Desktop App

Once your card passes the live nozzle checks and camera simulation, you can export directly to production-ready multi-material geometry ready for slicer ingestion.

> **Two STL Production Mode (Recommended)**
>
> Set the Export dropdown to **"Two STLs (plate + inlay) for 2 colours"** and click **Export STL**. The generator invokes OpenSCAD in the background using the Manifold 3D engine, producing two aligned models: `card_1_inlay.stl` and `card_2_plate.stl`. If OpenSCAD is not installed, use **Save .SCAD** or **Copy OpenSCAD** to render manually.

Bambu Studio Slicing & Printing Setup

1. **Simultaneous Import:** Open Bambu Studio. Drag and drop *both* `card_1_inlay.stl` and `card_2_plate.stl` onto the build plate at the exact same moment.
2. **Load as One Multi-Part Object:** Bambu Studio will prompt: *"Multiple objects were detected, do you want to load these files as a single object with multiple parts?"* Click **YES**. This locks their relative coordinate origins.
3. **Critical Filament Slot Order:** In the left Objects hierarchy, assign:
   - **Filament 1 (Lower Number)** → **`_1_inlay`** (e.g., White or Cyan)
   - **Filament 2 (Higher Number)** → **`_2_plate`** (e.g., Black or Charcoal)
4. **Why this saves time:** Bambu printers execute the first layer starting with the lowest-numbered loaded filament. The inlay prints first at Z=0. The printer then purges once to switch to Filament 2, fills the plate base around the letters, and immediately continues into Layer 2 and beyond without any further filament changes.
5. **Layer Height Calibration:** In your process settings, ensure **First Layer Height is set to 0.20 mm** to match the exact mathematical inlay depth defined in the model.
6. **Build Plate Selection:** A textured PEI sheet produces a scratch-resistant matte finish that diffuses room reflections and ensures camera scanning from any angle.

> **Verification Checklist:** Slice the plate and inspect the Layer 1 preview in Bambu Studio. You should see text/QR lines rendered in Colour A with zero empty perimeters, enclosed by Colour B. Total color switches for the print should equal exactly 1.